

Wanting Xiang

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<https://xiang-wt.github.io/>

EDUCATION

Xi'an Jiaotong University (C9), School of Mathematics and Statistics Xi'an, China

- Master of Applied Statistics, GPA: 3.50/4, Average: 87.80 Sept. 2023 – June 2026

Zhengzhou University (211), School of Business Zhengzhou, China

- Bachelor of Economic Statistics, GPA: 3.74/4, Average: 89.57 Sept. 2019 – June 2023

PUBLICATIONS

Development and Validation of an Interpretable Machine Learning Model for Predicting Tic Disorders and Severity in Children Based on Electroencephalogram Data

- Journal: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (JCR Q1, IF = 5.2)
- Coauthors: **Wanting Xiang (first author)**, Gang Zhu, Yichong Hou et al.
- Publication Date: June 2025
- DOI: 10.1109/TNSRE.2025.3579763

RESEARCH EXPERIENCE

Multi-branch GAT for Predicting TD Based on EEG Microstates Mar. 2025 – Present

- Proposed a Mi-STTF feature extraction method based on functional connectivity networks, combining microstate-driven dynamic connectivity with network topology
- Developed an interpretable MulB-GAT framework that adopted an intra-branch learning and inter-branch attention-based fusion
- Achieved 95.5% accuracy and identified YGTSS-related prefrontal-temporal abnormalities
- Manuscript Under Review

Two-Stage Progressive Diagnostic Design for Predicting TD Feb. 2024 – June 2025

- Preprocessed EEG from TD and HC using automated pipelines with manual refinement
- Proposed individual-based feature-weighted integration method and SFSW strategy, enhancing prediction accuracy of TD diagnosis and severity using ML models
- Employed model interpretability to uncover potential neurophysiological biomarkers
- Published one paper in 2025 *IEEE TNSRE*

Automatic Localization Design for Pharyngeal Sampling Regions Dec. 2022 – June 2023

- Proposed application of Mask R-CNN model to novel clinical scenario-automatic detection of pharyngeal regions for throat swab collection
- Optimized Mask R-CNN parameters for precise localization of pharyngeal sampling regions
- Documented in undergraduate dissertation

RELATED EXPERIENCE

Conferences

- The 7th International Conference on Mathematical Biology 2024, Xi'an, China
- Frontier Forum on Mathematical Methods for Pediatric Brain Disorders 2023, Xi'an, China

Work

- NIO Inc. | Data Operations Intern Feb. – May 2023
- Henan Public Opinion Research Center | Data Analysis Intern July – Aug. 2021

HONORS & AWARDS

- National Scholarship, Top 3% 2025
- “Huawei Cup” Mathematical Contest in Modeling, National Third Prize 2023, 2024
- Mathematical Contest in Modeling (MCM/ICM), Meritorious Winner 2022
- “Challenge Cup” Academic and Scientific Works Competition, National Second Prize 2022

SKILLS

Languages: Chinese (native), English (IELTS: 6.5)

Programming: Python, MATLAB, R, SPSS, SQL